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ORIGINAL ARTICLE

Acid leaching of mixed spent Li-ion batteries

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KEYWORDS

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Abstract Acid leaching for different types of mixed spent Li-ion mobile batteries is carried out after alkali decomposition using NH_4OH followed by H_2SO_4 + H_2O_2 leaching. In the alkali decomposition step, the effects of reaction time, NH_4OH concentration, liquid/solid mass ratio and reaction temperature on the decomposition process are investigated to remove Al, Cu, Mn, Ni, Co, and Li. After alkaline treatment, the alkali paste is treated to leach the remaining metals using H_2SO_4 + H_2O_2. The significant effects of reaction time, acid concentration, H_2O_2 concentration, liquid/solid mass ratios and reaction temperature on the leaching rate are studied. More than 97% of Al, Mn, Ni, Co, and Li and about 65% Cu are leached in two stages. Kinetic analysis shows that, the data fit with chemical reaction control mechanism and the activation energies for the investigated metals using the Arrhenius equation ranged from 30.1 to 41.4 kJ/mol. Recovered metals are precipitated from the leaching liquor at varying pH values using NaOH solution and Na_2CO_3. Firstly, Mn is precipitated as $MnCO_3$ at pH = 7.5. Secondly, at pH = 9.0, nickel is precipitated as $NiCO_3$. Thirdly, as the pH of the leaching liquor reaches 11–12, $Co(OH)_2$ is precipitated and the remaining Li is readily precipitated as Li_2CO_3 using a saturated Na_2CO_3 solution. Based on the experimental data, a flow sheet is developed and tested for the recovery process.

1. Introduction

In the last decades, the demands on charged batteries used in different electronic devices, such as cellular phones, are growing rapidly worldwide (Manis et al., 2013). The discarded spent cellular phone batteries cause serious environmental problems because it contains relatively high concentration of hazardous metals in their electrodes. Various types of batteries, such as; Li-ion (LIBs), nickel–cadmium (Ni–Cd), and nickel-metal hydride (Ni-MH) batteries, are used for different electronic products. Due to its good performance compared to the other batteries, LIBs are becoming the most dominant powerful source (Wang et al., 2014; Zeng et al., 2012; Chen et al., 2011; Kang et al., 2010a,b). These batteries are composed of a cathode, an anode, an electrolyte, and a separator (Manis et al., 2013). Thus, the production of LIBs and consequently, discarded waste is increased dramatically. For

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example, the worldwide production of LIB unit was nearly 2044 million in 2007 (Scrosati et al., 2007), and was up to about 4.6 billion unit in 2010 (Zeng et al., 2012).

Spent LIBs are defined as hazardous waste. If not handled properly, it will cause very serious harmful effect to the environment, animals and human health. On the other hand, there are so much valued metals in the spent LIBs, like Co, Li, Mn, ... etc. (Shin et al., 2005). Therefore, recycling of these spent batteries is necessary and important from both economical aspect (Kang et al., 2010a,b) as well as environmental protection. Recycling processes make economic sense where the recovered materials are chemically important, quite valuable, and to avoid disposal costs. Furthermore, the metal value in spent LIBs when recovered, represents an important secondary source for these metals with a higher grade than those found in natural minerals and ores.

Different pyrometallurgical and hydrometallurgical processes were investigated to leach and recover such valuable metal components from spent LIBs (Zhao et al., 2011; Chen et al., 2011; Pranolo et al., 2010; Kang et al., 2010a,b; Mantuano et al., 2006) In the literature, different hydrometallurgical techniques are used to leach and recover the metals ions from spent LIBs (Joulié et al., 2014; Lupi and Pasquali, 2003; Contestabile et al., 1999). The recovery processes are generally carried out by using ammoniacal and/or acidic leaching processes, precipitation, and thermal processes (Senanayake et al., 2010; Sayilgan et al., 2010; Xiao et al., 2009; Peng et al., 2008; Freitas et al., 2007; De Michelis et al., 2007).

Different leaching processes of LIBs were carried out using inorganic acids as leaching agents, such as H_2SO_4 (Nan et al., 2006; Swain et al., 2007; Shin et al., 2005; Mantuano et al., 2006), HCl (Wang et al., 2009; Li et al., 2009a,b; Zhang et al., 1998; Contestabile et al., 2001), and HNO_3 (Ferreira et al., 2009; Castillo et al., 2002; Lee and Rhee, 2002; Lee and Rhee, 2003). H_2O_2 is usually added in order to convert cobalt ions to the $+2$ state.

Till now, the recycling of different types of mixed spent LIBs used in cellular phones is limited in the literature. Thus, it is important to develop and test a simple and environmentally acceptable recycling process to leach and recover the valuable metals possible. Therefore, in this work, the decomposition of nine different types of spent Li-ion phone batteries by 4.0 M NH_4OH is investigated followed by leaching the resulted alkali paste by 2.0 M H_2SO_4 and 4.0% H_2O_2. In this context, different factors affecting leaching of Al, Cu, Mn, Ni, Co, and Li are investigated and assessed. The data obtained are validated by testing to fit a kinetic model and characterized the process. Accordingly Mn, Ni, Co, and Li are precipitated from the leach liquors by adjusting the pH value with NaOH and/or Na_2CO_3. A flow sheet is developed and tested to recover base metals from spent mixed type LIBs.

2. Experimental

2.1. Material

Different types of spent LIBs, (BL-5B, BP-4L, BL-5CB, BL-5CA, BL-6F, BL-5C, BL-4C, BL-4U, and BR-5C) were dismantled using a manual procedure described elsewhere (Dorella and Mansur, 2007; Mantuano et al., 2006) to remove both plastic and steel cases that cover the batteries. Once dismantled, anode and cathode were crushed carefully and sieved to separate the scrap paper, plastic film, outer metallic body, and membrane. Then, the black powder obtained was collected and washed with water to remove entrained electrolyte. The resulted powder was dried for 24 h at 60 °C, sieved with screens of 0.5, 2.0, and 5.0 mm.

In order to characterize the metal content in the powder of spent LIBs, Table 1, a sample of powdered materials that cover anode and cathode foils (Carbon corresponds to 16–18 wt.% of the overall sample) are dried and analyzed by X-ray fluorescence (XRF spectrometer, Asios, Sequential WD-XRF Spectrometer, PANalytical 2005) at National Research Centre, Egypt. Lithium concentration is analyzed by ICP-OES (Perkin Elmer Optima 2000 DV) at the Central Metallurgy R&D Institute, Helwan, Egypt.

The quantitative metal contents for Al, Mn, Cu, Ni, Co, and Li were assessed by atomic absorption spectrometry (AA spectrometer, GBC 932 plus model) after dissolving samples (anode and cathode, including metallic foils) in the leaching solutions.

2.2. Experimental procedure

All the experiments were conducted in batches, and the dissolution experiments were performed in 500-ml conical flasks. After adding known amounts of the mixed battery powder sample with NH_4OH solution, the mixture was heated under specific conditions. The decomposition conditions were fixed at 4.0 M NH_4OH solution at 60 °C for 60 min with a liquid/solid mass ratio of 15/1 except otherwise cited. At the end of each decomposition process, the slurry (paste) was filtered, washed with distilled water, and was taken for further acid leaching investigations.

The effect of agitation on the decomposition and leaching processes was studied using different stirring rates varying from 125 to 425 rpm. The results obtained shows that the dissolution rate increases with increasing stirring rate, but becomes almost independent of stirring above 250 rpm. Therefore, all experiments were performed at a stirring rate of 250 rpm to ensure good mass transfer.

3. Results and discussion

3.1. Decomposition with NH_4OH

In this section, decomposition processes are carried out for the raw material by using different concentrations of NH_4OH

Table 1 Typical metal composition of spent mixed LIBs powder (wt.%).

Compounds	wt.%	Compounds	wt.%
SiO_2	0.11	MgO	0.07
Al_2O_3	0.46	P_2O_5	1.39
CaO	0.22	Cr_2O_3	0.25
TiO_2	0.04	SO_3	0.27
Na_2O	0.08	CuO	4.11
$Fe_2O_3^{total}$	0.25	Li	2.67
ZrO_2	0.01	MnO	24.94
Co_2O_3	46.72	Cl	0.01
NiO	0.10	F	0.75

solution. The particles finer than 0.5 mm (gives better recovery efficiency) was treated by digesting with NH_4OH and the alkali paste was dried at 80 °C for 120 min. The main parameters that influence this process are decomposition time, NH_4OH concentration, NH_4OH-to-raw material (L/S) mass ratio and temperatures.

Dissolution of Al, Mn, Cu, Ni, Co, and Li was studied in the decomposition process. Recovery percentage (%E) of Al was increased very fast to reach about 97% after 50 min, while the recovery percentage (%E) of Cu, Mn, Ni, Co, and Li was slightly increased with time, reaching a plateau after about 60 min by using 4.0 M NH_4OH with L/S mass ratio of 15/1 at 60 °C (97.8% Al, 64.7% Cu). Therefore, all remaining tests were carried out at a constant time of 60 min.

The effect of NH_4OH concentration on dissolution of Al, Cu, Mn, Ni, Co, and Li from powder materials of LIBs was studied for 60 min with L/S mass ratio of 15/1 at 60 °C as shown in Fig. 1a. (%E) of Al increased from 20.5% to 99%, and (%E) of Cu also increased from 6.5% to 61.2% when the concentration of NH_4OH was increased from 0.5 M to 6.0 M. (%E) of Li and Ni increased from 4.6% to 14.8%, and from 1.6% to 4.7%, respectively, as NH_4OH concentration was increased from 0.5 M to 6.0 M. (%E) of Mn and Co (%E < 1.0%) by NH_4OH solutions is slightly affected, and their behaviors are nearly similar under these conditions. At 4.0 M NH_4OH, a plateau is formed with high (%E) values for Al and Cu (97.8% Al and 64.7% Cu). Therefore, all remaining experiments were carried out by 4.0 M NH_4OH solution.

As shown in Fig. 1b, the effect of temperature on the recovery percentage (%E) of the investigated elements was studied by using 4.0 M NH_4OH with L/S mass ratio of 15/1 for 60 min in the range of 25–80 °C. It can be observed that the dissolution efficiency of Al and Cu increases as the temperature increases until 60 °C. Moreover, as the temperature increases, small amounts (%E < 1.0%) of Mn and Co (their behaviors are nearly similar), about 11% Li and 4.5% Ni are recovered in the solution.

Generally, the decomposition of spent LIBs by NH_4OH was found as a very important and interesting step to be included in a treatment route for spent LIBs because it is very selective for Al and Cu over the other metal content. The alkali paste obtained is washed with water in order to remove the impurities and then dried at 80 °C for 120 min. The resulted paste contains the main bulk of Mn, Co, Ni, and Li. This paste was subjected to the following acid leaching investigations.

3.2. Leaching study

3.2.1. Effect of leaching time

To study the effect of leaching time on leaching process of the alkali paste of spent LIBs by 2.0 M H_2SO_4, several experiments were carried out at a time from 15 to 180 min. The other dissolution parameters were fixed with an L/S mass ratio of 10/1, 4.0% H_2O_2 at 70 °C. The results obtained, Fig. 1c, shows that the leaching efficiency (%E) of Co reached 92.8% at first 45 min and a plateau is obtained after 60 min with (%E) of 99.1%. Leaching percentage was increased to 66.5% for Mn, 57.7% for Ni, and to 59.7 for Li, respectively, when the time was increased to 60 min. Furthermore, as the time increases to 120 °C, (%E) of Mn, Ni, and Li increases to the maximum values. Further increase in leaching time from 120 to 180 min has a slight effect on the dissolution rate of Mn, Ni, and Li and adverse effect on the dissolution rate of Co. Therefore, all remaining tests were carried out at a constant leaching time of 120 min.

3.2.2. Effect of a liquid/solid mass ratio

Effect of L/S mass ratio on leaching efficiency (%E) of Mn, Ni, Co, and Li from the alkali residue of spent LIBs was studied using 2.0 M H_2SO_4 acid solution with reaction time of

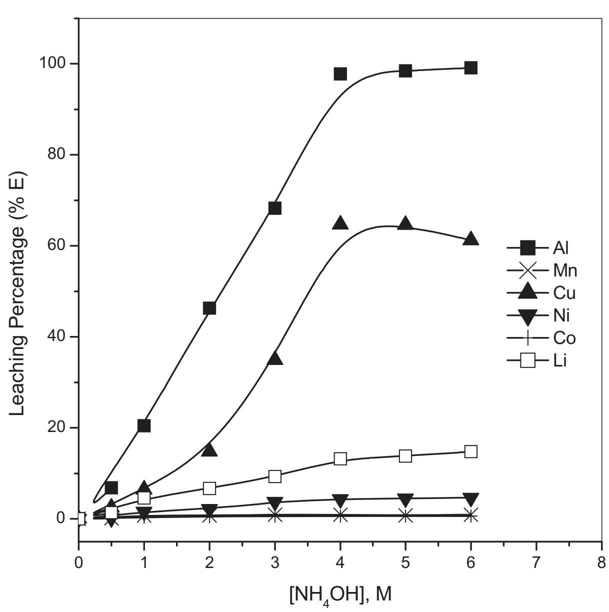

Figure 1a Effect of NH_4OH concentration on the decomposition rate of spent LiBs (contact time = 60 min, L/S mass ratio = 15/1, T, °C = 60 °C).

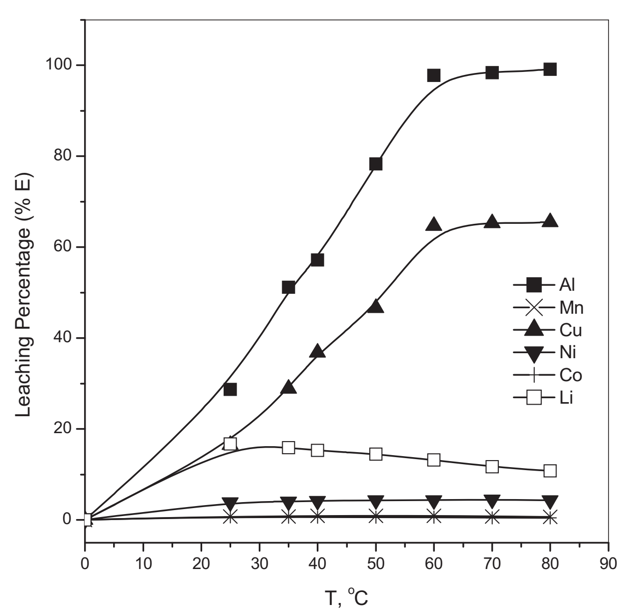

Figure 1b Effect of temperature on the decomposition rate of spent LiBs by 4.0 M NH_4OH (contact time = 60 min, L/S mass ratio = 15/1).

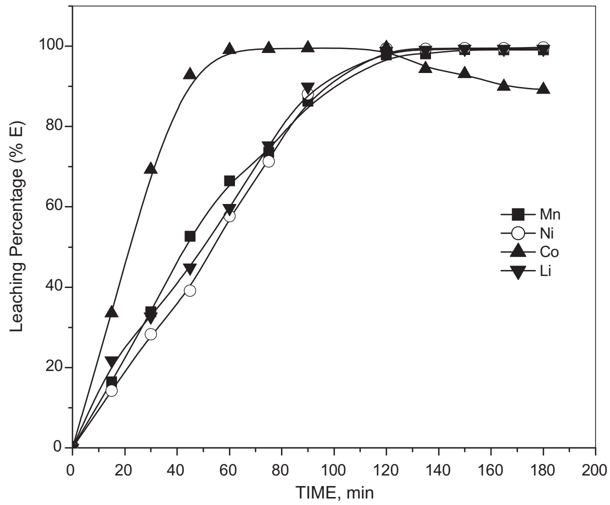

Figure 1c Effect of time on the leaching of alkali residue of spent LiBs by 2.0 M H_2SO_4 (H_2O_2 = 4.0%, L/S mass ratio = 10/1, T, °C = 70 °C).

120 min and 4.0% H_2O_2 at 70 °C, Fig. 2a. The results obtained indicate that (%E) was increased to 97.8% for Mn, 99.4% for Ni, 99.6% for Co and 98.8% for Li, respectively, as the liquid/alkali paste mass ratio increases from 1/1 to 10/1. When the L/S mass ratio increases from 10/1 to 15/1, (%E) of Mn slightly increased, while no significant change in the leaching percent of Ni, Co, and Li was observed. The leaching of Mn, Ni, Co, and Li decreased when the L/S mass ratio was increased from 15/1 to 20/1.

The increase in leaching efficiency of the investigated elements when the liquid/alkali paste mass ratio increased can be attributed to the decrease in the viscosity and consequently, decreases the mass transfer resistance in the acid–alkali paste of the spent LIB interface. Further, at a low L/S mass ratio there is insufficient acid to react with the alkali paste whereby at a certain higher L/S mass ratio more acid is available to react with it to give maximum recovery of Mn, Ni, Co, and Li. Thus, the best condition of L/S ratio was 10/1.

3.2.3. Effect of H_2SO_4 concentration

Several leaching experiments were performed using H_2SO_4 solutions with concentrations varying from 0.5 to 4.0 M for 120 min with 4.0% H_2O_2 and L/S mass ratio of 10/1 at 70 °C as shown in Fig. 2b. The results obtained show that (%E) of Ni increases rapidly to 87.8% as the H_2SO_4 concentration increases to 1.0 M while (%E) of Mn, Co, and Li slightly increased in this range of concentration. Using higher concentration of H_2SO_4 up to 2.0 M, (%E) of the investigated metals rapidly increased up to 97.8% for Mn, 99.4% for Ni, 99.6% for Co and 98.8% for Li. With further increase in the H_2SO_4 concentration from 2.0 to 4.0 M, (%E) of the investigated metals increased slowly in comparison to lower H_2SO_4 concentration solutions in the presence and absence of H_2O_2.

This behavior can be explained by the following chemical equations of dissolution of alkali paste of spent LIBs (such as $LiCoO_2$ and $LiMnO_2$ types) in the H_2SO_4solution in the absence and in the presence of H_2O_2 solution (Nan et al., 2005; Zhu et al., 2012).

$$4LiCoO_{2(s)} + 6H_2SO_{4(aq)} \rightarrow 4CoSO_{4(aq)} + 2Li_2SO_{4(aq)} + 6H_2O_{(g)} + O_{2(g)} \quad (1a)$$

$$2LiCoO_{2(s)} + 3H_2SO_{4(aq)} + H_2O_{2(aq)} \rightarrow 2CoSO_{4(aq)} + Li_2SO_{4(aq)} + 4H_2O_{(g)} + O_{2(g)} \quad (1b)$$

$$4LiMnO_{2(s)} + 6H_2SO_{4(aq)} \rightarrow 4MnSO_{4(aq)} + 2Li_2SO4(aq) + 6H_2O_{(g)} + O_{2(g)} \quad (2a)$$

$$2LiMnO_{2(s)} + 3H_2SO_{4(aq)} + H_2O_{2(aq)} \rightarrow 2MnSO_{4(aq)} + Li_2SO_{4(aq)} + 4H_2O_{(g)} + O_{2(g)} \quad (2b)$$

These equations show that the addition of the reacting substances can facilitate the forward reaction resulting in an increase in (%E) of the investigated metals.

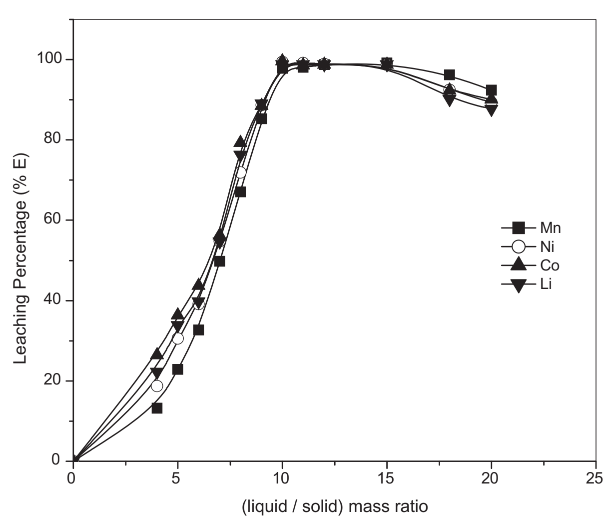

Figure 2a Effect of L/S mass ratio on the leaching of alkali residue of spent LiBs by 2.0 M H_2SO_4 (H_2O_2 = 4.0%, time = 120 min, T, °C = 70 °C).

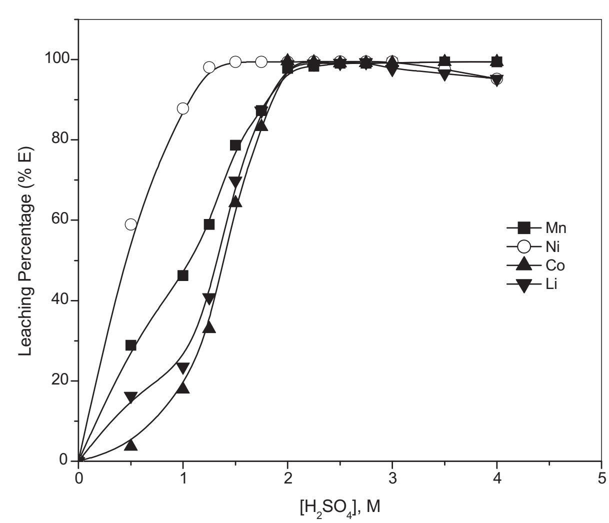

Figure 2b Effect of H_2SO_4 concentration on the leaching of alkali residue of spent LiBs. (H_2O_2 = 4.0%, time = 120 min, L/S mass ratio = 10/1, T, °C = 70 °C).

3.2.4. Effect of H_2O_2 concentration

Effect of concentration of H_2O_2 on the leaching of alkali paste of spent LIBs by 2.0 M H_2SO_4 was studied in the range from 0.0 to 6.0%, Fig. 2c. The results obtained show only 51.6% of Mn and about 42.7% of Co were leached in the absence of H_2O_2 solution, whereas more than 82.2% of Ni and about 74.5% of Li were leached at the same conditions. (%*E*) of Mn and Co was increased significantly with the increase in concentration of H_2O_2 from 0.0% to 4.0% to reach about 97.8% of Mn and 99.6% of Co and (%*E*) of Ni and Li increased from 82.2% to 99.4% and 74.5% to 98.8%, respectively, under the investigated conditions.

Fig. 2c shows that (%*E*) did not increase significantly when more than 4.0% H_2O_2 was used. This can be due to the fact that is instability of H_2O_2 solutions. When heated, H_2O_2 solution can be decomposed according to the following equation (Eq. (3));

$$H_2O_{2(aq)} \rightarrow H_2O_{(aq)} + 1/2O_{2(g)} \quad (3)$$

Therefore, an increase in H_2O_2 concentration could accelerate its decomposition, resulting in no significant increase in the leaching efficiency of alkali paste of spent LIBs by 2.0 H_2SO_4 solution at high concentration of H_2O_2 (above 4.0%).

3.2.5. Effect of temperature

Effect of leaching temperature on the dissolution of alkali paste of LIBs by 2.0 M H_2SO_4 was studied in the range of 20–100 °C, while the other leaching parameters remained constant. The results obtained are shown in Fig. 2d. It can be observed that (%*E*) of Mn, Ni, Co, and Li increases with the increase in temperature. The data obtained illustrate that leaching efficiency is significantly affected by temperature. The results indicate that only 52.7% Mn, 23.8% Ni, 29.7% Co, and 18.6% Li are leached at 20 °C. At 70 °C, about 97.8% Mn, 99.4% Ni, 99.6% Co, and 98.8% Li was recovered. A further increase in the temperature does not show any significant increase in the recovery of metals. Thus, the best condition of temperature is 70 °C. This indicates that the leaching process of metals from spent LIBs is an endothermic reaction (Sakuitung et al., 2007).

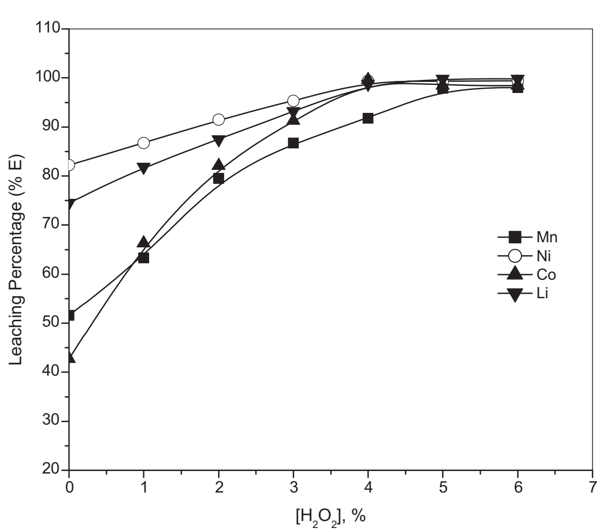

Figure 2c Effect of H_2O_2 concentration on the leaching of alkali residue of spent LiBs by 2.0 M H_2SO_4 (time = 120 min, *L*/*S* mass ratio = 10/1, *T*, °C = 70 °C).

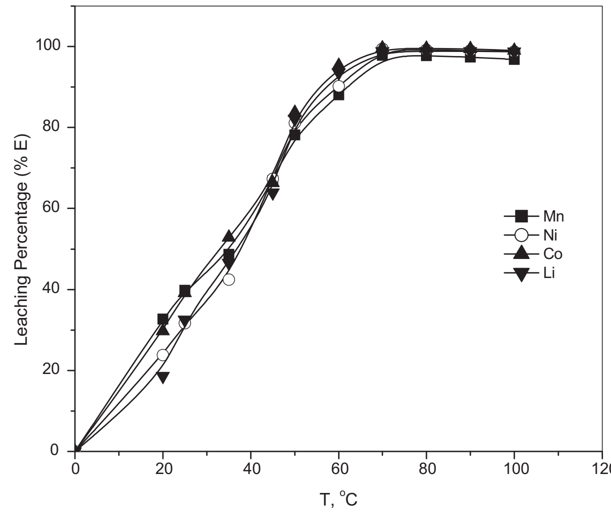

Figure 2d Effect of temperature on the leaching of alkali residue of spent LiBs by 2.0 M H_2SO_4 (H_2O_2 = 4.0%, time = 120 min, *L*/*S* mass ratio = 10/1).

3.3. Kinetic analysis

In order to determine the kinetic parameters and rate controlling step in the leaching of the investigated elements (Mn, Ni, Co, and Li) from the alkali residue of spent LIBs by 2.0 M H_2SO_4 in the presence of 4.0% H_2O_2, the experimental data in Fig. 2d, are processed and correlated to various kinetic models for solid–liquid reactions. The equations of the shrinking core model when either the surface chemical reactions or diffusion reactions are the slowest step can be expressed as follows, respectively, (Levenspiel, 1998).

$$1-(1-X)^{1/3}=k_c t \quad (4)$$

$$1-2/3X-(1-X)^{2/3}=k_d t \quad (5)$$

where X is the fraction reacted, k_c is the chemical reaction rate constant (min^{-1}), k_d is the apparent diffusion reaction rate constant (min^{-1}), and t is the leaching time (min).

The results obtained for Mn, Ni, Co, and Li, presented in Fig. 3a–d as a relation between $1-(1-X)^{1/3}$ against time (t), gave straight lines at different temperatures and pass through the origin. This verifies Eq. (4). Under these conditions, the reactions are controlled by surface chemical reactions. The correlation coefficients of these straight lines are about 0.99.

For estimation of the activation energy, the reaction between the overall rate constant and temperature from the Eq. (4) can be expressed by the Arrhenius equation, as follows;

$$k = Ae^{E_a/T} \quad (6)$$

where k is the reaction rate constant, A is the frequency factor, E_a is the apparent activation energy and R is the gas constant ($=8.314472\ J\ k^{-1}\ mol^{-1}$).

The rate constants for different temperatures at different times were calculated from Fig. 3a–d for the investigated elements. A plot of ln k vs. $1/T$ for this model is linear (Fig. 4) and the values of the apparent activation energy (E_a) for the leaching elements were determined from the line slopes. The

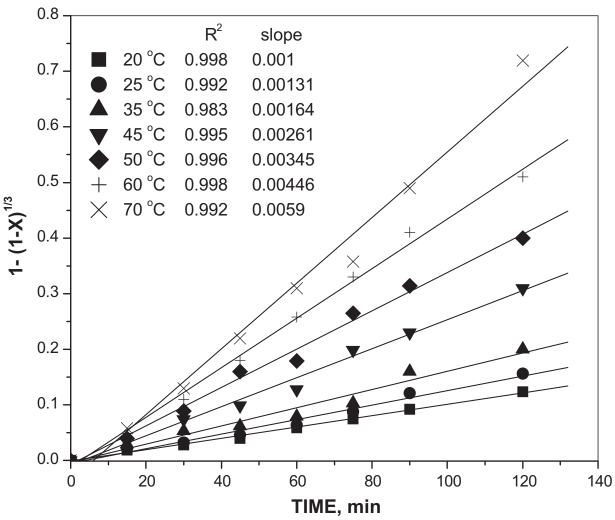

Figure 3a Plots of $1-(1-X)^{1/3}$ vs. time for Mn leaching at various temperatures by 2.0 M H_2SO_4 solution.

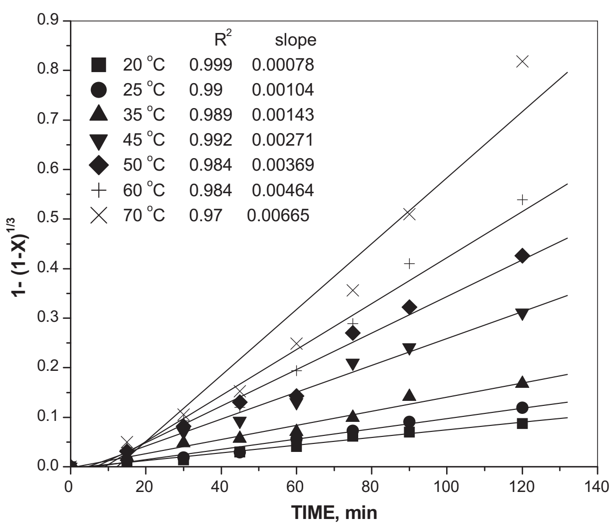

Figure 3b Plots of $1-(1-X)^{1/3}$ vs. time for Ni leaching at various temperatures by 2.0 M H_2SO_4 solution.

calculated values are 30.1 kJ/mol for Mn, 36.7 kJ/mol for Ni, 41.4 kJ/mol for Co, and 37.4 kJ/mol for Li.

The relatively high values of activation energies obtained for leaching of these elements by H_2SO_4 in the presence of H_2O_2 is clearly indicative of a chemical controlled reaction in this leaching process.

3.4. Recovery of leached metals from leaching liquor of spent LIBs

Based on the results obtained, a flow sheet for the separation of Mn, Ni, Co, and Li is developed and tested. After the decomposition process of LIBs by 4.0 M NH_4OH, the resulted alkali paste has been leached by 2.0 M H_2SO_4 and 4.0% H_2O_2. The resulted leach liquor (1000 ml) was found to contain 32.73 g Co, 18.7 g Mn, 2.265 g Li, 0.075 g Ni, and other impurities, Fig. 5.

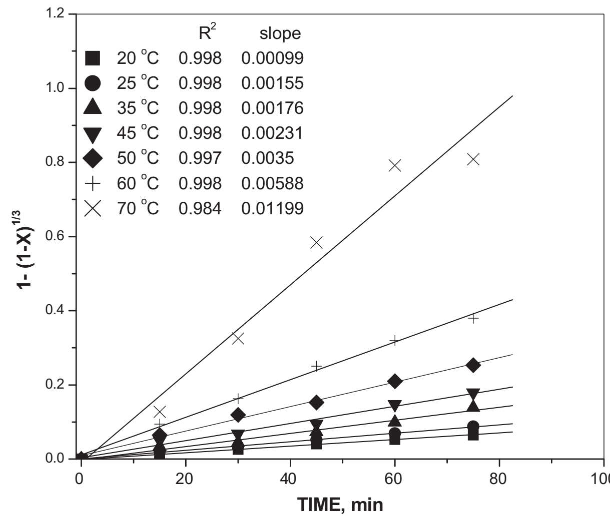

Figure 3c Plots of $1-(1-X)^{1/3}$ vs. time for Co leaching at various temperatures by 2.0 M H_2SO_4 solution.

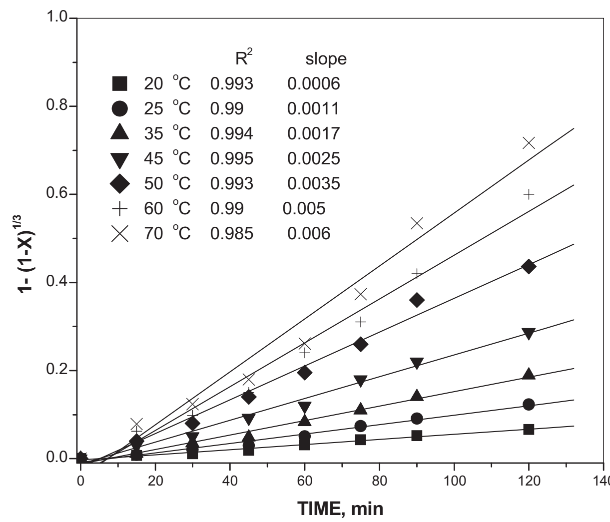

Figure 3d Plots of $1-(1-X)^{1/3}$ vs. time for Li leaching at various temperatures by 2.0 M H_2SO_4 solution.

Leach liquor was, first, purified by filtration to remove any suspended material. By adjusting the pH value of the leach liquor to 7.5 by adding 2.0 M NaOH solution into the filtration, Mn was precipitated with saturated solution of Na_2CO_3. White to faint pink precipitate of $MnCO_3$ is obtained and washed using hot water to remove the soluble salts. The resulting $MnCO_3$ is dried at 100 °C for 1.0 h. However, the analytical results show that about 94% of Mn is recovered as a precipitate.

The pH value of leach liquor was adjusted by 2.0 M NaOH to 9.0. With adding a solution of Na_2CO_3 and stirring for 1.0 h at room temperature, nickel is precipitated as light green precipitate ($NiCO_3$). Solid precipitate was washed with water to remove the soluble salt and then dried at 60 °C for about 4.0 h. At these conditions, about 91% of Ni is recovered from the leaching liquor.

A saturated solution of NaOH was added to the leach liquor under stirring to reach a pH of 11–12. After 2.0 h under

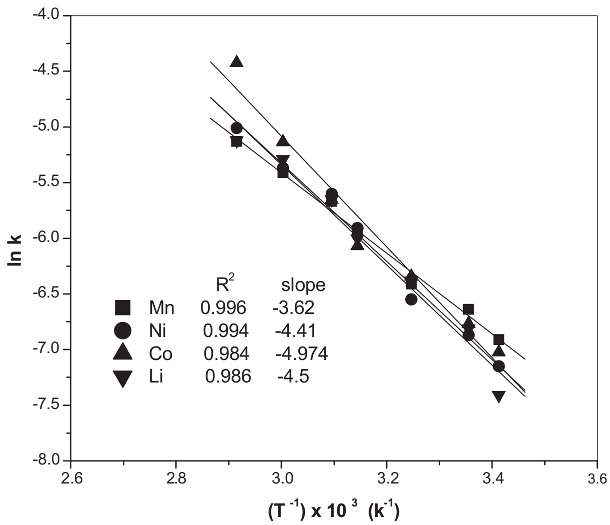

Figure 4 Arrhenius plot for Mn, Ni, Co, and Li leaching in 2.0 M H_2SO_4.

stirring at room temperature, the rose red precipitate $Co(OH)_2$ is separated by filtration. Then, solid precipitates were washed with warm water to remove different soluble salts and then dried at 100 °C for about 4.0 h. About 95% of Co is precipitated from the leaching solution under these optimum conditions.

By adding the Na_2CO_3 solution to the filtrate, the solution was agitated at a speed of 250 rpm for 1.0 h to precipitate Li_2CO_3. The white precipitate obtained is separated from the leach liquor by filtration. The precipitate obtained is washed using hot water to eliminate the entrained sodium and then dried at 100 °C for 1.0 h. The data obtained show that about 90% of Li is recovered as a precipitate.

4. Conclusions

In this work, a procedure has been established for dissolving and recovery Al, Cu, Mn, Co, Ni, and Li existing in the powder resulted from crushing, and mixing of different types of spent LIBs. Therefore, we have studied the effect of decomposition and leaching processes on recovery of these metals. In the decomposition process, 97.8% Al and 64.7% Cu have been leached by using 4.0 M NH_4OH with a L/S mass ratio of 15/1 at 60 °C for 60 min. Then, about 97.8% Mn, 99.4% Ni, 99.6% Co, and 98.8% Li have been leached from the alkali residue of spent LIBs using 2.0 M H_2SO_4 with 4.0% H_2O_2 for 120 min with a L/S mass ratio of 10/1 at 70 °C. Kinetic studies showed that, the data fit with chemical reaction control mechanism and the activation energies for the investigated metals are 30.1 kJ/mol for Mn, 36.7 kJ/mol for Ni, 41.4 kJ/mol for Co, and 37.4 kJ/mol for Li. Based on the experimental results, a flow sheet for the recovery process has been developed and tested to separate Mn, Ni, Co, and Li from LIBs.

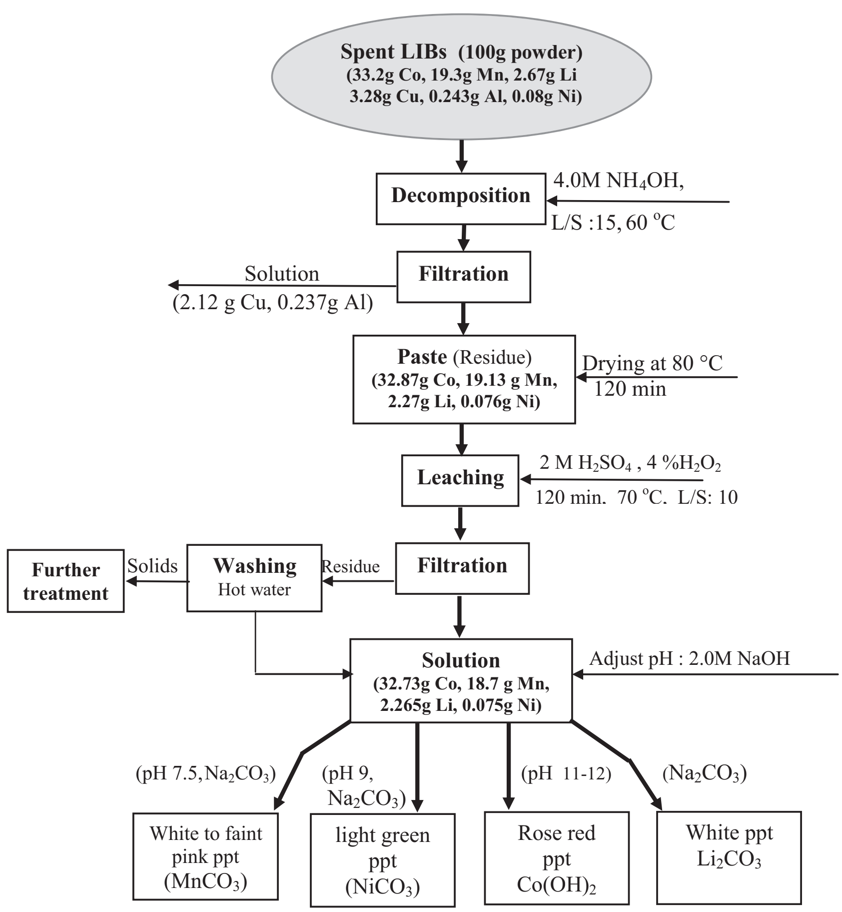

Figure 5 Flow sheet for the recovery of Mn(II), Ni(II), Co(II), and Li(II) from spent LIBs.

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